NetMind Study Guide

Project charter and success measures

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Project Manager, John Arvie Capulo should be ready to explain how this part of NetMind supports scope, delivery, risks, stakeholder communication, approvals, and release readiness.

System knowledge

NetMind is a volleyball analytics and team-management system. It keeps users, teams, memberships, trainer assignments, seasons, matches, drills, training assignments, performance logs, wellness check-ins, attendance, injuries, lineups, rankings, comparisons, notifications, and push subscriptions.

Database relationships and constraints protect integrity: user roles and volleyball positions are restricted, team membership is unique, check-ins are unique per player and date, attendance statuses are controlled, and records use foreign keys to connect a player, staff member, team, season, or match.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Project Manager must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Stakeholder map and role boundaries

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System knowledge

The system has a React/Vite web client, an Expo mobile client, and a Node.js/Express API. The web client contains coach and trainer dashboards; the mobile client includes dashboard, activity, attendance, performance, training, wellness, injury, notifications, profile, and settings views.

A player summary combines performance actions, wellness indicators, check-ins, training completion, drill results, and active injuries. A team dashboard combines team information, match record, attendance, and rankings. These summaries support decisions but remain dependent on accurate inputs.

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NetMind Study Guide

End-to-end system map

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System knowledge

The API uses a protected /api area. After a Bearer JWT is validated, authorization checks determine whether the authenticated coach, trainer, or player can access the requested team, player, match, or record. Interface visibility is not a substitute for server authorization.

Training flows begin with a drill or assigned focus, target values, and a due date. Players and staff record status and drill-performance information. The records then support follow-up, completion tracking, and performance analysis.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
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NetMind Study Guide

Release planning and work breakdown

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System knowledge

Authentication validates inputs, hashes passwords with bcrypt, signs JWT tokens, rate-limits registration and login, and rechecks the active user account when a token is used. This protects account access and blocks deactivated users.

Attendance can be recorded in bulk and is tied to team, player, session type, date, status, staff marker, and optional late minutes or reason. Training schedules and QR tokens support controlled check-in; token expiry and authorization are important exception cases.

Flow to explain during the presentation

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NetMind Study Guide

Requirements traceability

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System knowledge

Database relationships and constraints protect integrity: user roles and volleyball positions are restricted, team membership is unique, check-ins are unique per player and date, attendance statuses are controlled, and records use foreign keys to connect a player, staff member, team, season, or match.

Match flows include a team, opponent, date, match type, season, scores, outcome, and optional per-match scoring. A finalized result can recalculate season ranking fields such as wins, losses, win percentage, points for, points against, and point difference.

Flow to explain during the presentation

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NetMind Study Guide

Risk register and mitigation

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System knowledge

A player summary combines performance actions, wellness indicators, check-ins, training completion, drill results, and active injuries. A team dashboard combines team information, match record, attendance, and rankings. These summaries support decisions but remain dependent on accurate inputs.

Sensitive personal and sports-health data must be handled with least privilege. Environment secrets, passwords, API keys, email settings, and Drive credentials must never appear in public documentation, screenshots, commits, test reports, or presentation slides.

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NetMind Study Guide

Access and data quality governance

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System knowledge

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NetMind Study Guide

Training and attendance workflow oversight

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System knowledge

Attendance can be recorded in bulk and is tied to team, player, session type, date, status, staff marker, and optional late minutes or reason. Training schedules and QR tokens support controlled check-in; token expiry and authorization are important exception cases.

The system has a React/Vite web client, an Expo mobile client, and a Node.js/Express API. The web client contains coach and trainer dashboards; the mobile client includes dashboard, activity, attendance, performance, training, wellness, injury, notifications, profile, and settings views.

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NetMind Study Guide

Match, wellness, and injury workflow oversight

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System knowledge

Match flows include a team, opponent, date, match type, season, scores, outcome, and optional per-match scoring. A finalized result can recalculate season ranking fields such as wins, losses, win percentage, points for, points against, and point difference.

The API uses a protected /api area. After a Bearer JWT is validated, authorization checks determine whether the authenticated coach, trainer, or player can access the requested team, player, match, or record. Interface visibility is not a substitute for server authorization.

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NetMind Study Guide

Handover and retrospective

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NetMind Study Guide

Repository orientation and startup

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NetMind Study Guide

Web frontend routing

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NetMind Study Guide

Mobile application structure

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System knowledge

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Training flows begin with a drill or assigned focus, target values, and a due date. Players and staff record status and drill-performance information. The records then support follow-up, completion tracking, and performance analysis.

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NetMind Study Guide

API request conventions

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NetMind Study Guide

Express application pipeline

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NetMind Study Guide

JWT authentication

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5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Programmer must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Authorization rules

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Programmer, Vincent Torres should be ready to explain how this part of NetMind supports implementation, debugging, integration, security, and deployment of the application.

System knowledge

Training flows begin with a drill or assigned focus, target values, and a due date. Players and staff record status and drill-performance information. The records then support follow-up, completion tracking, and performance analysis.

NetMind is a volleyball analytics and team-management system. It keeps users, teams, memberships, trainer assignments, seasons, matches, drills, training assignments, performance logs, wellness check-ins, attendance, injuries, lineups, rankings, comparisons, notifications, and push subscriptions.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Programmer must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
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Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Database schema

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Programmer, Vincent Torres should be ready to explain how this part of NetMind supports implementation, debugging, integration, security, and deployment of the application.

System knowledge

Attendance can be recorded in bulk and is tied to team, player, session type, date, status, staff marker, and optional late minutes or reason. Training schedules and QR tokens support controlled check-in; token expiry and authorization are important exception cases.

The system has a React/Vite web client, an Expo mobile client, and a Node.js/Express API. The web client contains coach and trainer dashboards; the mobile client includes dashboard, activity, attendance, performance, training, wellness, injury, notifications, profile, and settings views.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
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4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Programmer must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

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Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Generic CRUD resources

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Programmer, Vincent Torres should be ready to explain how this part of NetMind supports implementation, debugging, integration, security, and deployment of the application.

System knowledge

Match flows include a team, opponent, date, match type, season, scores, outcome, and optional per-match scoring. A finalized result can recalculate season ranking fields such as wins, losses, win percentage, points for, points against, and point difference.

The API uses a protected /api area. After a Bearer JWT is validated, authorization checks determine whether the authenticated coach, trainer, or player can access the requested team, player, match, or record. Interface visibility is not a substitute for server authorization.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Programmer must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
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Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Team and roster implementation

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Programmer, Vincent Torres should be ready to explain how this part of NetMind supports implementation, debugging, integration, security, and deployment of the application.

System knowledge

Sensitive personal and sports-health data must be handled with least privilege. Environment secrets, passwords, API keys, email settings, and Drive credentials must never appear in public documentation, screenshots, commits, test reports, or presentation slides.

Authentication validates inputs, hashes passwords with bcrypt, signs JWT tokens, rate-limits registration and login, and rechecks the active user account when a token is used. This protects account access and blocks deactivated users.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Programmer must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Trainer assignment implementation

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Programmer, Vincent Torres should be ready to explain how this part of NetMind supports implementation, debugging, integration, security, and deployment of the application.

System knowledge

NetMind is a volleyball analytics and team-management system. It keeps users, teams, memberships, trainer assignments, seasons, matches, drills, training assignments, performance logs, wellness check-ins, attendance, injuries, lineups, rankings, comparisons, notifications, and push subscriptions.

Database relationships and constraints protect integrity: user roles and volleyball positions are restricted, team membership is unique, check-ins are unique per player and date, attendance statuses are controlled, and records use foreign keys to connect a player, staff member, team, season, or match.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Programmer must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Drills and training assignments

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Programmer, Vincent Torres should be ready to explain how this part of NetMind supports implementation, debugging, integration, security, and deployment of the application.

System knowledge

The system has a React/Vite web client, an Expo mobile client, and a Node.js/Express API. The web client contains coach and trainer dashboards; the mobile client includes dashboard, activity, attendance, performance, training, wellness, injury, notifications, profile, and settings views.

A player summary combines performance actions, wellness indicators, check-ins, training completion, drill results, and active injuries. A team dashboard combines team information, match record, attendance, and rankings. These summaries support decisions but remain dependent on accurate inputs.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Programmer must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

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- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
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Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Drill performance logs

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Programmer, Vincent Torres should be ready to explain how this part of NetMind supports implementation, debugging, integration, security, and deployment of the application.

System knowledge

The API uses a protected /api area. After a Bearer JWT is validated, authorization checks determine whether the authenticated coach, trainer, or player can access the requested team, player, match, or record. Interface visibility is not a substitute for server authorization.

Training flows begin with a drill or assigned focus, target values, and a due date. Players and staff record status and drill-performance information. The records then support follow-up, completion tracking, and performance analysis.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
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Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Performance logs and summaries

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Programmer, Vincent Torres should be ready to explain how this part of NetMind supports implementation, debugging, integration, security, and deployment of the application.

System knowledge

Authentication validates inputs, hashes passwords with bcrypt, signs JWT tokens, rate-limits registration and login, and rechecks the active user account when a token is used. This protects account access and blocks deactivated users.

Attendance can be recorded in bulk and is tied to team, player, session type, date, status, staff marker, and optional late minutes or reason. Training schedules and QR tokens support controlled check-in; token expiry and authorization are important exception cases.

Flow to explain during the presentation

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2. Identify the client screen or mobile action and the API request it requires.
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Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Wellness and injury features

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Programmer, Vincent Torres should be ready to explain how this part of NetMind supports implementation, debugging, integration, security, and deployment of the application.

System knowledge

Database relationships and constraints protect integrity: user roles and volleyball positions are restricted, team membership is unique, check-ins are unique per player and date, attendance statuses are controlled, and records use foreign keys to connect a player, staff member, team, season, or match.

Match flows include a team, opponent, date, match type, season, scores, outcome, and optional per-match scoring. A finalized result can recalculate season ranking fields such as wins, losses, win percentage, points for, points against, and point difference.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
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Presentation checkpoint

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NetMind Study Guide

Schedules, QR, and attendance

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Programmer, Vincent Torres should be ready to explain how this part of NetMind supports implementation, debugging, integration, security, and deployment of the application.

System knowledge

A player summary combines performance actions, wellness indicators, check-ins, training completion, drill results, and active injuries. A team dashboard combines team information, match record, attendance, and rankings. These summaries support decisions but remain dependent on accurate inputs.

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NetMind Study Guide

Matches, lineups, and scores

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Programmer, Vincent Torres should be ready to explain how this part of NetMind supports implementation, debugging, integration, security, and deployment of the application.

System knowledge

Training flows begin with a drill or assigned focus, target values, and a due date. Players and staff record status and drill-performance information. The records then support follow-up, completion tracking, and performance analysis.

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Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
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Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Rankings and analytics

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Programmer, Vincent Torres should be ready to explain how this part of NetMind supports implementation, debugging, integration, security, and deployment of the application.

System knowledge

Attendance can be recorded in bulk and is tied to team, player, session type, date, status, staff marker, and optional late minutes or reason. Training schedules and QR tokens support controlled check-in; token expiry and authorization are important exception cases.

The system has a React/Vite web client, an Expo mobile client, and a Node.js/Express API. The web client contains coach and trainer dashboards; the mobile client includes dashboard, activity, attendance, performance, training, wellness, injury, notifications, profile, and settings views.

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Study questions

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Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Files, notifications, and email

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Programmer, Vincent Torres should be ready to explain how this part of NetMind supports implementation, debugging, integration, security, and deployment of the application.

System knowledge

Match flows include a team, opponent, date, match type, season, scores, outcome, and optional per-match scoring. A finalized result can recalculate season ranking fields such as wins, losses, win percentage, points for, points against, and point difference.

The API uses a protected /api area. After a Bearer JWT is validated, authorization checks determine whether the authenticated coach, trainer, or player can access the requested team, player, match, or record. Interface visibility is not a substitute for server authorization.

Flow to explain during the presentation

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Study questions

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NetMind Study Guide

Testing and deployment

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Programmer, Vincent Torres should be ready to explain how this part of NetMind supports implementation, debugging, integration, security, and deployment of the application.

System knowledge

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Authentication validates inputs, hashes passwords with bcrypt, signs JWT tokens, rate-limits registration and login, and rechecks the active user account when a token is used. This protects account access and blocks deactivated users.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
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Responsibilities for this role

A Programmer must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
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Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Test strategy and system inventory

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Tester, Ervin Augis should be ready to explain how this part of NetMind supports functional, permission, reliability, data-integrity, and release testing.

System knowledge

NetMind is a volleyball analytics and team-management system. It keeps users, teams, memberships, trainer assignments, seasons, matches, drills, training assignments, performance logs, wellness check-ins, attendance, injuries, lineups, rankings, comparisons, notifications, and push subscriptions.

Database relationships and constraints protect integrity: user roles and volleyball positions are restricted, team membership is unique, check-ins are unique per player and date, attendance statuses are controlled, and records use foreign keys to connect a player, staff member, team, season, or match.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Tester must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
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Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Test data and environments

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Tester, Ervin Augis should be ready to explain how this part of NetMind supports functional, permission, reliability, data-integrity, and release testing.

System knowledge

The system has a React/Vite web client, an Expo mobile client, and a Node.js/Express API. The web client contains coach and trainer dashboards; the mobile client includes dashboard, activity, attendance, performance, training, wellness, injury, notifications, profile, and settings views.

A player summary combines performance actions, wellness indicators, check-ins, training completion, drill results, and active injuries. A team dashboard combines team information, match record, attendance, and rankings. These summaries support decisions but remain dependent on accurate inputs.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Tester must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Registration and login tests

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Tester, Ervin Augis should be ready to explain how this part of NetMind supports functional, permission, reliability, data-integrity, and release testing.

System knowledge

The API uses a protected /api area. After a Bearer JWT is validated, authorization checks determine whether the authenticated coach, trainer, or player can access the requested team, player, match, or record. Interface visibility is not a substitute for server authorization.

Training flows begin with a drill or assigned focus, target values, and a due date. Players and staff record status and drill-performance information. The records then support follow-up, completion tracking, and performance analysis.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Tester must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Session and authorization tests

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Tester, Ervin Augis should be ready to explain how this part of NetMind supports functional, permission, reliability, data-integrity, and release testing.

System knowledge

Authentication validates inputs, hashes passwords with bcrypt, signs JWT tokens, rate-limits registration and login, and rechecks the active user account when a token is used. This protects account access and blocks deactivated users.

Attendance can be recorded in bulk and is tied to team, player, session type, date, status, staff marker, and optional late minutes or reason. Training schedules and QR tokens support controlled check-in; token expiry and authorization are important exception cases.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Tester must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Team and roster tests

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Tester, Ervin Augis should be ready to explain how this part of NetMind supports functional, permission, reliability, data-integrity, and release testing.

System knowledge

Database relationships and constraints protect integrity: user roles and volleyball positions are restricted, team membership is unique, check-ins are unique per player and date, attendance statuses are controlled, and records use foreign keys to connect a player, staff member, team, season, or match.

Match flows include a team, opponent, date, match type, season, scores, outcome, and optional per-match scoring. A finalized result can recalculate season ranking fields such as wins, losses, win percentage, points for, points against, and point difference.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Tester must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Training and performance tests

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Tester, Ervin Augis should be ready to explain how this part of NetMind supports functional, permission, reliability, data-integrity, and release testing.

System knowledge

A player summary combines performance actions, wellness indicators, check-ins, training completion, drill results, and active injuries. A team dashboard combines team information, match record, attendance, and rankings. These summaries support decisions but remain dependent on accurate inputs.

Sensitive personal and sports-health data must be handled with least privilege. Environment secrets, passwords, API keys, email settings, and Drive credentials must never appear in public documentation, screenshots, commits, test reports, or presentation slides.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Tester must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Attendance and QR tests

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Tester, Ervin Augis should be ready to explain how this part of NetMind supports functional, permission, reliability, data-integrity, and release testing.

System knowledge

Training flows begin with a drill or assigned focus, target values, and a due date. Players and staff record status and drill-performance information. The records then support follow-up, completion tracking, and performance analysis.

NetMind is a volleyball analytics and team-management system. It keeps users, teams, memberships, trainer assignments, seasons, matches, drills, training assignments, performance logs, wellness check-ins, attendance, injuries, lineups, rankings, comparisons, notifications, and push subscriptions.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Tester must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Match, lineup, and ranking tests

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Tester, Ervin Augis should be ready to explain how this part of NetMind supports functional, permission, reliability, data-integrity, and release testing.

System knowledge

Attendance can be recorded in bulk and is tied to team, player, session type, date, status, staff marker, and optional late minutes or reason. Training schedules and QR tokens support controlled check-in; token expiry and authorization are important exception cases.

The system has a React/Vite web client, an Expo mobile client, and a Node.js/Express API. The web client contains coach and trainer dashboards; the mobile client includes dashboard, activity, attendance, performance, training, wellness, injury, notifications, profile, and settings views.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Tester must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Wellness, injury, and notification tests

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Tester, Ervin Augis should be ready to explain how this part of NetMind supports functional, permission, reliability, data-integrity, and release testing.

System knowledge

Match flows include a team, opponent, date, match type, season, scores, outcome, and optional per-match scoring. A finalized result can recalculate season ranking fields such as wins, losses, win percentage, points for, points against, and point difference.

The API uses a protected /api area. After a Bearer JWT is validated, authorization checks determine whether the authenticated coach, trainer, or player can access the requested team, player, match, or record. Interface visibility is not a substitute for server authorization.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Tester must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Regression report and defect handover

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Tester, Ervin Augis should be ready to explain how this part of NetMind supports functional, permission, reliability, data-integrity, and release testing.

System knowledge

Sensitive personal and sports-health data must be handled with least privilege. Environment secrets, passwords, API keys, email settings, and Drive credentials must never appear in public documentation, screenshots, commits, test reports, or presentation slides.

Authentication validates inputs, hashes passwords with bcrypt, signs JWT tokens, rate-limits registration and login, and rechecks the active user account when a token is used. This protects account access and blocks deactivated users.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Tester must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
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Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Documentation plan and audience

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Documentary, Althea Adobas should be ready to explain how this part of NetMind supports accurate user, technical, operational, and presentation documentation.

System knowledge

NetMind is a volleyball analytics and team-management system. It keeps users, teams, memberships, trainer assignments, seasons, matches, drills, training assignments, performance logs, wellness check-ins, attendance, injuries, lineups, rankings, comparisons, notifications, and push subscriptions.

Database relationships and constraints protect integrity: user roles and volleyball positions are restricted, team membership is unique, check-ins are unique per player and date, attendance statuses are controlled, and records use foreign keys to connect a player, staff member, team, season, or match.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Documentary must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
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- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

System overview and glossary

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Documentary, Althea Adobas should be ready to explain how this part of NetMind supports accurate user, technical, operational, and presentation documentation.

System knowledge

The system has a React/Vite web client, an Expo mobile client, and a Node.js/Express API. The web client contains coach and trainer dashboards; the mobile client includes dashboard, activity, attendance, performance, training, wellness, injury, notifications, profile, and settings views.

A player summary combines performance actions, wellness indicators, check-ins, training completion, drill results, and active injuries. A team dashboard combines team information, match record, attendance, and rankings. These summaries support decisions but remain dependent on accurate inputs.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Documentary must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Architecture narrative

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Documentary, Althea Adobas should be ready to explain how this part of NetMind supports accurate user, technical, operational, and presentation documentation.

System knowledge

The API uses a protected /api area. After a Bearer JWT is validated, authorization checks determine whether the authenticated coach, trainer, or player can access the requested team, player, match, or record. Interface visibility is not a substitute for server authorization.

Training flows begin with a drill or assigned focus, target values, and a due date. Players and staff record status and drill-performance information. The records then support follow-up, completion tracking, and performance analysis.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Documentary must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
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- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Roles and permissions guide

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Documentary, Althea Adobas should be ready to explain how this part of NetMind supports accurate user, technical, operational, and presentation documentation.

System knowledge

Authentication validates inputs, hashes passwords with bcrypt, signs JWT tokens, rate-limits registration and login, and rechecks the active user account when a token is used. This protects account access and blocks deactivated users.

Attendance can be recorded in bulk and is tied to team, player, session type, date, status, staff marker, and optional late minutes or reason. Training schedules and QR tokens support controlled check-in; token expiry and authorization are important exception cases.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Documentary must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Account lifecycle guide

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Documentary, Althea Adobas should be ready to explain how this part of NetMind supports accurate user, technical, operational, and presentation documentation.

System knowledge

Database relationships and constraints protect integrity: user roles and volleyball positions are restricted, team membership is unique, check-ins are unique per player and date, attendance statuses are controlled, and records use foreign keys to connect a player, staff member, team, season, or match.

Match flows include a team, opponent, date, match type, season, scores, outcome, and optional per-match scoring. A finalized result can recalculate season ranking fields such as wins, losses, win percentage, points for, points against, and point difference.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Documentary must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Team and training guide

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Documentary, Althea Adobas should be ready to explain how this part of NetMind supports accurate user, technical, operational, and presentation documentation.

System knowledge

A player summary combines performance actions, wellness indicators, check-ins, training completion, drill results, and active injuries. A team dashboard combines team information, match record, attendance, and rankings. These summaries support decisions but remain dependent on accurate inputs.

Sensitive personal and sports-health data must be handled with least privilege. Environment secrets, passwords, API keys, email settings, and Drive credentials must never appear in public documentation, screenshots, commits, test reports, or presentation slides.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Documentary must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Performance and analytics guide

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Documentary, Althea Adobas should be ready to explain how this part of NetMind supports accurate user, technical, operational, and presentation documentation.

System knowledge

Training flows begin with a drill or assigned focus, target values, and a due date. Players and staff record status and drill-performance information. The records then support follow-up, completion tracking, and performance analysis.

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Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A Documentary must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Match, attendance, wellness, and injury guide

> **Role objective.** Study this topic to prepare for the capstone presentation. As the Documentary, Althea Adobas should be ready to explain how this part of NetMind supports accurate user, technical, operational, and presentation documentation.

System knowledge

Attendance can be recorded in bulk and is tied to team, player, session type, date, status, staff marker, and optional late minutes or reason. Training schedules and QR tokens support controlled check-in; token expiry and authorization are important exception cases.

The system has a React/Vite web client, an Expo mobile client, and a Node.js/Express API. The web client contains coach and trainer dashboards; the mobile client includes dashboard, activity, attendance, performance, training, wellness, injury, notifications, profile, and settings views.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
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NetMind Study Guide

API, data, configuration, and operations reference

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System knowledge

Match flows include a team, opponent, date, match type, season, scores, outcome, and optional per-match scoring. A finalized result can recalculate season ranking fields such as wins, losses, win percentage, points for, points against, and point difference.

The API uses a protected /api area. After a Bearer JWT is validated, authorization checks determine whether the authenticated coach, trainer, or player can access the requested team, player, match, or record. Interface visibility is not a substitute for server authorization.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
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NetMind Study Guide

Review, versioning, and final package

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System knowledge

Sensitive personal and sports-health data must be handled with least privilege. Environment secrets, passwords, API keys, email settings, and Drive credentials must never appear in public documentation, screenshots, commits, test reports, or presentation slides.

Authentication validates inputs, hashes passwords with bcrypt, signs JWT tokens, rate-limits registration and login, and rechecks the active user account when a token is used. This protects account access and blocks deactivated users.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
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NetMind Study Guide

Problem and scope

> **Role objective.** Study this topic to prepare for the capstone presentation. As the System Analyst, Mae Kyla Olita should be ready to explain how this part of NetMind supports requirements, use cases, rules, data definitions, and acceptance criteria.

System knowledge

NetMind is a volleyball analytics and team-management system. It keeps users, teams, memberships, trainer assignments, seasons, matches, drills, training assignments, performance logs, wellness check-ins, attendance, injuries, lineups, rankings, comparisons, notifications, and push subscriptions.

Database relationships and constraints protect integrity: user roles and volleyball positions are restricted, team membership is unique, check-ins are unique per player and date, attendance statuses are controlled, and records use foreign keys to connect a player, staff member, team, season, or match.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
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Responsibilities for this role

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NetMind Study Guide

Actors and use cases

> **Role objective.** Study this topic to prepare for the capstone presentation. As the System Analyst, Mae Kyla Olita should be ready to explain how this part of NetMind supports requirements, use cases, rules, data definitions, and acceptance criteria.

System knowledge

The system has a React/Vite web client, an Expo mobile client, and a Node.js/Express API. The web client contains coach and trainer dashboards; the mobile client includes dashboard, activity, attendance, performance, training, wellness, injury, notifications, profile, and settings views.

A player summary combines performance actions, wellness indicators, check-ins, training completion, drill results, and active injuries. A team dashboard combines team information, match record, attendance, and rankings. These summaries support decisions but remain dependent on accurate inputs.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
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NetMind Study Guide

Context and boundary analysis

> **Role objective.** Study this topic to prepare for the capstone presentation. As the System Analyst, Mae Kyla Olita should be ready to explain how this part of NetMind supports requirements, use cases, rules, data definitions, and acceptance criteria.

System knowledge

The API uses a protected /api area. After a Bearer JWT is validated, authorization checks determine whether the authenticated coach, trainer, or player can access the requested team, player, match, or record. Interface visibility is not a substitute for server authorization.

Training flows begin with a drill or assigned focus, target values, and a due date. Players and staff record status and drill-performance information. The records then support follow-up, completion tracking, and performance analysis.

Flow to explain during the presentation

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Responsibilities for this role

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NetMind Study Guide

Permission matrix

> **Role objective.** Study this topic to prepare for the capstone presentation. As the System Analyst, Mae Kyla Olita should be ready to explain how this part of NetMind supports requirements, use cases, rules, data definitions, and acceptance criteria.

System knowledge

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Attendance can be recorded in bulk and is tied to team, player, session type, date, status, staff marker, and optional late minutes or reason. Training schedules and QR tokens support controlled check-in; token expiry and authorization are important exception cases.

Flow to explain during the presentation

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NetMind Study Guide

Data model

> **Role objective.** Study this topic to prepare for the capstone presentation. As the System Analyst, Mae Kyla Olita should be ready to explain how this part of NetMind supports requirements, use cases, rules, data definitions, and acceptance criteria.

System knowledge

Database relationships and constraints protect integrity: user roles and volleyball positions are restricted, team membership is unique, check-ins are unique per player and date, attendance statuses are controlled, and records use foreign keys to connect a player, staff member, team, season, or match.

Match flows include a team, opponent, date, match type, season, scores, outcome, and optional per-match scoring. A finalized result can recalculate season ranking fields such as wins, losses, win percentage, points for, points against, and point difference.

Flow to explain during the presentation

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NetMind Study Guide

Authentication requirements

> **Role objective.** Study this topic to prepare for the capstone presentation. As the System Analyst, Mae Kyla Olita should be ready to explain how this part of NetMind supports requirements, use cases, rules, data definitions, and acceptance criteria.

System knowledge

A player summary combines performance actions, wellness indicators, check-ins, training completion, drill results, and active injuries. A team dashboard combines team information, match record, attendance, and rankings. These summaries support decisions but remain dependent on accurate inputs.

Sensitive personal and sports-health data must be handled with least privilege. Environment secrets, passwords, API keys, email settings, and Drive credentials must never appear in public documentation, screenshots, commits, test reports, or presentation slides.

Flow to explain during the presentation

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NetMind Study Guide

Team and training requirements

> **Role objective.** Study this topic to prepare for the capstone presentation. As the System Analyst, Mae Kyla Olita should be ready to explain how this part of NetMind supports requirements, use cases, rules, data definitions, and acceptance criteria.

System knowledge

Training flows begin with a drill or assigned focus, target values, and a due date. Players and staff record status and drill-performance information. The records then support follow-up, completion tracking, and performance analysis.

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Flow to explain during the presentation

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Responsibilities for this role

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Study questions

- Which user role is allowed to start, review, or change this information?
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Presentation checkpoint

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NetMind Study Guide

Performance, wellness, and attendance requirements

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System knowledge

Attendance can be recorded in bulk and is tied to team, player, session type, date, status, staff marker, and optional late minutes or reason. Training schedules and QR tokens support controlled check-in; token expiry and authorization are important exception cases.

The system has a React/Vite web client, an Expo mobile client, and a Node.js/Express API. The web client contains coach and trainer dashboards; the mobile client includes dashboard, activity, attendance, performance, training, wellness, injury, notifications, profile, and settings views.

Flow to explain during the presentation

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NetMind Study Guide

Match, ranking, and analytics requirements

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System knowledge

Match flows include a team, opponent, date, match type, season, scores, outcome, and optional per-match scoring. A finalized result can recalculate season ranking fields such as wins, losses, win percentage, points for, points against, and point difference.

The API uses a protected /api area. After a Bearer JWT is validated, authorization checks determine whether the authenticated coach, trainer, or player can access the requested team, player, match, or record. Interface visibility is not a substitute for server authorization.

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NetMind Study Guide

Traceability and acceptance criteria

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System knowledge

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NetMind Study Guide

Current-to-future workflow

> **Role objective.** Study this topic to prepare for the capstone presentation. As the System Analyst, Rea Batiao should be ready to explain how this part of NetMind supports workflows, integrations, exceptions, measurable rules, and change impact analysis.

System knowledge

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Database relationships and constraints protect integrity: user roles and volleyball positions are restricted, team membership is unique, check-ins are unique per player and date, attendance statuses are controlled, and records use foreign keys to connect a player, staff member, team, season, or match.

Flow to explain during the presentation

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Responsibilities for this role

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NetMind Study Guide

Flow notation and system boundary

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System knowledge

The system has a React/Vite web client, an Expo mobile client, and a Node.js/Express API. The web client contains coach and trainer dashboards; the mobile client includes dashboard, activity, attendance, performance, training, wellness, injury, notifications, profile, and settings views.

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NetMind Study Guide

Account and authorization analysis

> **Role objective.** Study this topic to prepare for the capstone presentation. As the System Analyst, Rea Batiao should be ready to explain how this part of NetMind supports workflows, integrations, exceptions, measurable rules, and change impact analysis.

System knowledge

The API uses a protected /api area. After a Bearer JWT is validated, authorization checks determine whether the authenticated coach, trainer, or player can access the requested team, player, match, or record. Interface visibility is not a substitute for server authorization.

Training flows begin with a drill or assigned focus, target values, and a due date. Players and staff record status and drill-performance information. The records then support follow-up, completion tracking, and performance analysis.

Flow to explain during the presentation

1. State the business trigger and the actor who starts the process.
2. Identify the client screen or mobile action and the API request it requires.
3. Explain validation, authentication, and role or ownership authorization before a write occurs.
4. Name the database record or relationship that is created, updated, read, calculated, or protected.
5. Describe the visible result: dashboard update, status, summary, ranking, notification, or clear error.

Responsibilities for this role

A System Analyst must make this flow understandable, reliable, and verifiable within the team. Capture the purpose, actors, inputs, preconditions, expected result, exception cases, and evidence. Confirm behavior against the web or mobile screen, backend route, data schema, and available tests rather than relying only on a design assumption.

Study questions

- Which user role is allowed to start, review, or change this information?
- What data proves that the action happened, and which dashboard or summary uses it?
- What invalid input, missing relationship, expired session, or unauthorized request must be handled?
- How would the team demonstrate that the workflow protects accuracy, privacy, and accountability?

Presentation checkpoint

Prepare a short explanation in this order: **need - actor - action - validation - protected data update - result - benefit**. Use an example from volleyball operations, such as assigning a drill, logging attendance, recording match performance, reviewing wellness, or finalizing a ranking. Mention the exception path as well as the successful path.

NetMind Study Guide

Profile and media process

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System knowledge

Authentication validates inputs, hashes passwords with bcrypt, signs JWT tokens, rate-limits registration and login, and rechecks the active user account when a token is used. This protects account access and blocks deactivated users.

Attendance can be recorded in bulk and is tied to team, player, session type, date, status, staff marker, and optional late minutes or reason. Training schedules and QR tokens support controlled check-in; token expiry and authorization are important exception cases.

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NetMind Study Guide

Team and trainer lifecycle

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System knowledge

Database relationships and constraints protect integrity: user roles and volleyball positions are restricted, team membership is unique, check-ins are unique per player and date, attendance statuses are controlled, and records use foreign keys to connect a player, staff member, team, season, or match.

Match flows include a team, opponent, date, match type, season, scores, outcome, and optional per-match scoring. A finalized result can recalculate season ranking fields such as wins, losses, win percentage, points for, points against, and point difference.

Flow to explain during the presentation

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NetMind Study Guide

Training and drill-log process

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System knowledge

A player summary combines performance actions, wellness indicators, check-ins, training completion, drill results, and active injuries. A team dashboard combines team information, match record, attendance, and rankings. These summaries support decisions but remain dependent on accurate inputs.

Sensitive personal and sports-health data must be handled with least privilege. Environment secrets, passwords, API keys, email settings, and Drive credentials must never appear in public documentation, screenshots, commits, test reports, or presentation slides.

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NetMind Study Guide

Performance and summary process

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System knowledge

Training flows begin with a drill or assigned focus, target values, and a due date. Players and staff record status and drill-performance information. The records then support follow-up, completion tracking, and performance analysis.

NetMind is a volleyball analytics and team-management system. It keeps users, teams, memberships, trainer assignments, seasons, matches, drills, training assignments, performance logs, wellness check-ins, attendance, injuries, lineups, rankings, comparisons, notifications, and push subscriptions.

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NetMind Study Guide

Wellness, injury, QR, and attendance process

> **Role objective.** Study this topic to prepare for the capstone presentation. As the System Analyst, Rea Batiao should be ready to explain how this part of NetMind supports workflows, integrations, exceptions, measurable rules, and change impact analysis.

System knowledge

Attendance can be recorded in bulk and is tied to team, player, session type, date, status, staff marker, and optional late minutes or reason. Training schedules and QR tokens support controlled check-in; token expiry and authorization are important exception cases.

The system has a React/Vite web client, an Expo mobile client, and a Node.js/Express API. The web client contains coach and trainer dashboards; the mobile client includes dashboard, activity, attendance, performance, training, wellness, injury, notifications, profile, and settings views.

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NetMind Study Guide

Match, ranking, and notification process

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System knowledge

Match flows include a team, opponent, date, match type, season, scores, outcome, and optional per-match scoring. A finalized result can recalculate season ranking fields such as wins, losses, win percentage, points for, points against, and point difference.

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NetMind Study Guide

Data integrity, exceptions, and change impact

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